

Market Risk, Counterparty Risk, Haircuts and Margining in Repo Transactions

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Abstract

A repurchase agreement (repo) is often viewed as a secured lending transaction in which counterparty credit risk is mitigated by collateral. However, the magnitude of the counterparty exposure is fundamentally driven by the market risk of the underlying collateral. This note illustrates how market volatility translates into counterparty exposure over the life of a repo, how haircuts reduce this exposure, and why periodic margining is often a more effective risk mitigant than relying on a fixed initial haircut alone.

1 Introduction

A repo transaction can be interpreted as a secured loan. One party temporarily exchanges a security for cash and agrees to reverse the transaction at a later date. Although the resulting exposure is a counterparty credit exposure, its magnitude is determined by fluctuations in the market value of the underlying collateral. Consequently, market risk and counterparty risk are closely linked. If the collateral appreciates during the life of the repo and the counterparty defaults, the lender may incur a loss when attempting to replace the collateral in the market. The credit exposure therefore arises from counterparty default but is driven by changes in market prices. The purpose of this note is to illustrate this mechanism using a stylized simulation of a three-month repo on a Swedish five-year government bond.

2 Interest Rate Dynamics and Bond Pricing

The bond price is assumed to depend on a single risk factor, namely the five-year interest rate. The interest rate is simulated as a random walk,

$$dy_t = \sigma_y \sqrt{dt} Z_t, \quad (1)$$

where

$$Z_t \sim N(0, 1),$$

and σ_y denotes the annualized interest-rate volatility. Bond prices are approximated using a standard duration-based relationship,

$$P_t = P_0 (1 - D(y_t - y_0)), \quad (2)$$

where P_0 is the initial bond value, D is the modified duration, and y_0 is the initial yield. This approximation captures the first-order sensitivity of bond prices to yield changes and is sufficient for illustrating the relationship between market volatility and counterparty exposure.

3 Counterparty Exposure Without Protection

Suppose first that the repo does not involve any haircut or margining. If the counterparty defaults at time t , the lender must replace the bond at its prevailing market value. The resulting exposure is therefore

$$E_t = \max(P_t - P_0, 0). \quad (3)$$

Two commonly used measures of counterparty credit risk are expected exposure (EE) and potential future exposure (PFE). They are defined as

$$EE_t = \mathbb{E}[E_t], \quad (4)$$

and

$$PFE_\alpha(t) = Q_\alpha(E_t), \quad (5)$$

where $Q_\alpha(\cdot)$ denotes the α -quantile of the exposure distribution. In the simulations below, particular attention is given to PFE_{99} , which measures the 99th percentile of future exposure.

The exposure is only realized if a default occurs and therefore constitutes a credit exposure. Nevertheless, its size is entirely determined by the market value of the bond. The key property is that volatility accumulates over time. Since the interest-rate process follows a random walk,

$$\sigma_y(t) \propto \sqrt{t}, \quad (6)$$

and the same scaling property applies approximately to bond price volatility. As a consequence, expected exposure and potential future exposure tend to increase with repo maturity. Even if the lender intends to hold the bond to maturity, the possibility of counterparty default introduces a replacement-cost risk that depends on market prices.

4 The Effect of Haircuts

A common way to reduce counterparty exposure is to apply a haircut. Instead of lending the full market value of the bond, the lender applies an initial deduction

$$H = hP_0, \quad (7)$$

where h denotes the haircut percentage. The exposure becomes

$$E_t = \max(P_t - P_0 - H, 0). \quad (8)$$

The haircut creates an initial buffer against adverse market movements. As long as cumulative price changes remain smaller than the haircut, no credit exposure arises. However, the haircut remains fixed throughout the transaction while the volatility of the underlying risk factor continues to accumulate over time. This implies that a fixed haircut eventually becomes less effective as repo maturity increases. To illustrate this point, note that the standard deviation of the bond price over a horizon T may be approximated by

$$\sigma_P(T) \approx D\sigma_y\sqrt{T}. \quad (9)$$

A simple approximation to the 99% potential future exposure is therefore

$$PFE_{99}(T) \approx \max(2.33\sigma_P(T) - H, 0). \quad (10)$$

The relation shows that a haircut must be assessed relative to the volatility of the underlying collateral and the relevant risk horizon. A haircut that is sufficient for a short repo may become inadequate for a longer transaction simply because market volatility increases with time.

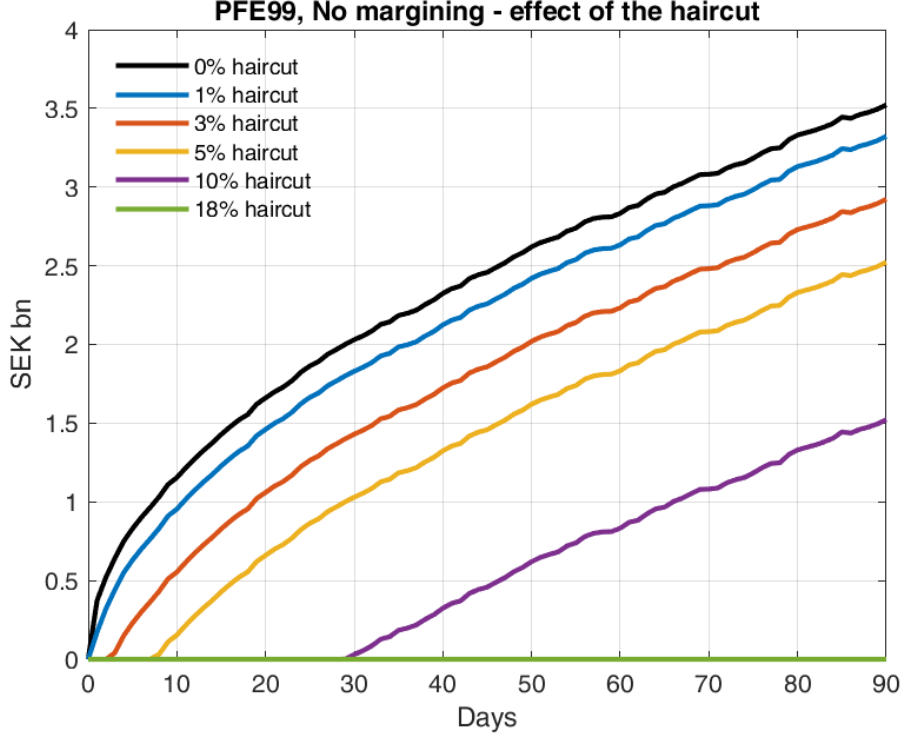


Figure 1: Potential future exposure at the 99th percentile for different haircut levels. A larger haircut reduces exposure by creating a larger initial buffer against adverse price movements. However, because volatility accumulates over time, the effectiveness of a fixed haircut declines as the risk horizon increases.

Figure 1 illustrates this effect. The exposure grows over time because the volatility of the underlying interest-rate process increases with the square root of time. Larger haircuts push the exposure profile downward, but a sufficiently long horizon will eventually erode the protection provided by any fixed haircut.

5 The Effect of Margining

An alternative and often more effective method of reducing exposure is periodic margining. Under a margining agreement, collateral is adjusted during the life of the repo. If margin calls occur every Δt days, the exposure becomes

$$E_t = \max(P_t - P_{t-\Delta t} - hP_{t-\Delta t}, 0). \quad (11)$$

The critical difference is that exposure is no longer measured relative to the initial bond value. Instead, exposure depends only on the price movement occurring between margin calls. This significantly shortens the effective risk horizon. Since volatility grows approximately with the square root of time, reducing the interval from three months to one month, one week, or one day dramatically reduces the amount of market risk that can accumulate before collateral is adjusted.

More generally, the relevant risk horizon in collateralized transactions is not necessarily the contractual maturity of the repo itself. In practice, exposure is typically assessed over the margin period of risk (MPOR), i.e. the period between the last successful margin call and the time required to close out or replace the position following a counterparty default. Frequent margining reduces the amount of market risk that can accumulate during the MPOR and therefore reduces counterparty exposure.

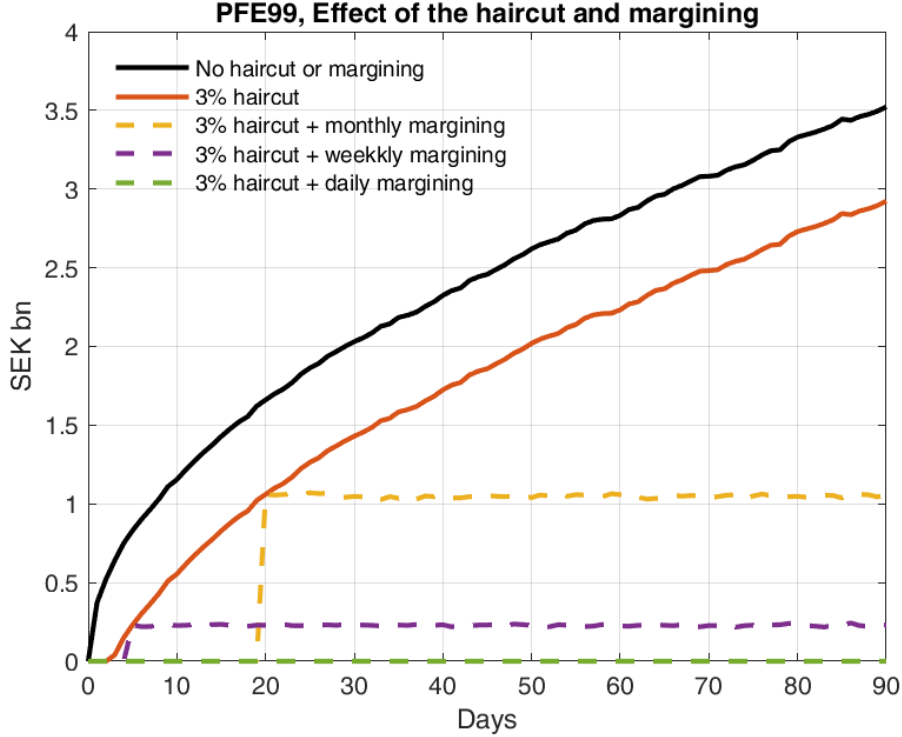


Figure 2: Potential future exposure at the 99th percentile under different margining frequencies. More frequent margining reduces the effective risk horizon and thereby limits the accumulation of market volatility between collateral adjustments.

Figure 2 shows that monthly, weekly, and daily margining all reduce exposure relative to a repo with a fixed initial haircut only. The simulations demonstrate that shortening the margin period can reduce potential future exposure more effectively than increasing the haircut by a moderate amount.

6 Conclusion

Repo exposures provide a clear example of how market risk can be transformed into counterparty credit risk. Without protection, exposure grows over time as volatility accumulates. Haircuts reduce exposure by providing an initial buffer, but their effectiveness depends on the volatility of the collateral and the relevant risk horizon. In practice, the relevant horizon is often determined by the margin period of risk (MPOR) rather than by the contractual maturity of the repo itself. For a well-margined repo, counterparty exposure is therefore driven primarily by market movements during the MPOR. Nevertheless, longer repo maturities may still be relevant from a liquidity-management and balance-sheet perspective, since collateral remains tied up for a longer period. Periodic margining reduces exposure by shortening the effective risk horizon. Haircuts and margining should therefore be viewed as complementary risk-management tools: haircuts provide protection against adverse price movements during the close-out period, while margining limits the accumulation of replacement-cost risk between collateral calls.